Lab 7: Fuzzing for Input Validation Bugs

Warranty Claims Application

Objective

Use property-based fuzz testing with Hypothesis to discover edge-case failures in parse_claim, age_average, and payable_charge, then harden the functions so malformed inputs do not crash the program.

Result summary

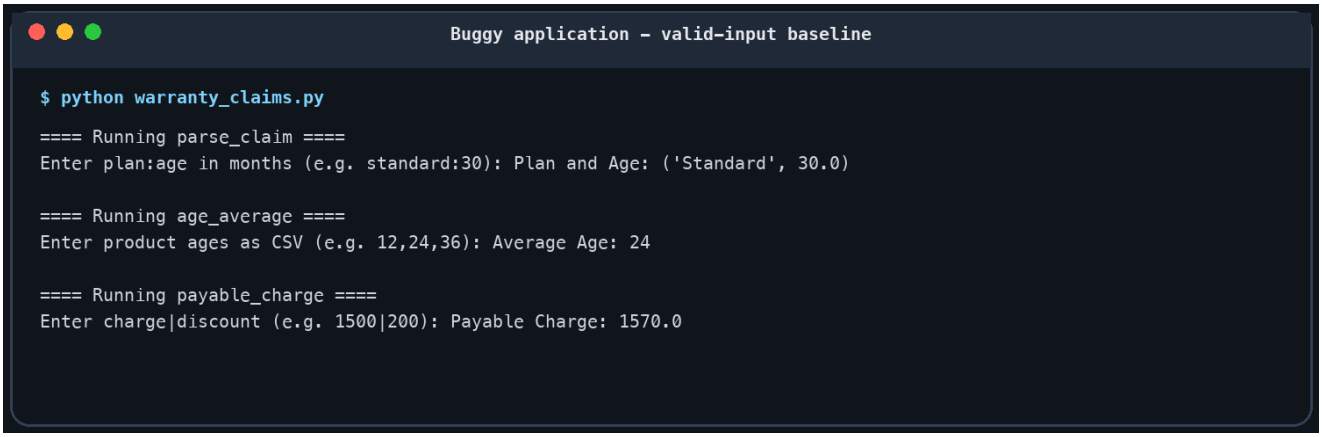
The initial fuzz run produced four failing tests and exposed ValueError and AttributeError paths for empty strings, missing delimiters, empty numeric fields, and None. The fixed implementation validates types, delimiters, numeric conversion, finiteness, and non-negative domain values. The final run passes all five tests.

Safe defaults

Invalid claims return ('Unknown', 0.0); invalid age lists return 0; invalid charge data returns 0.0. Valid-input behavior is preserved and checked with regression assertions.

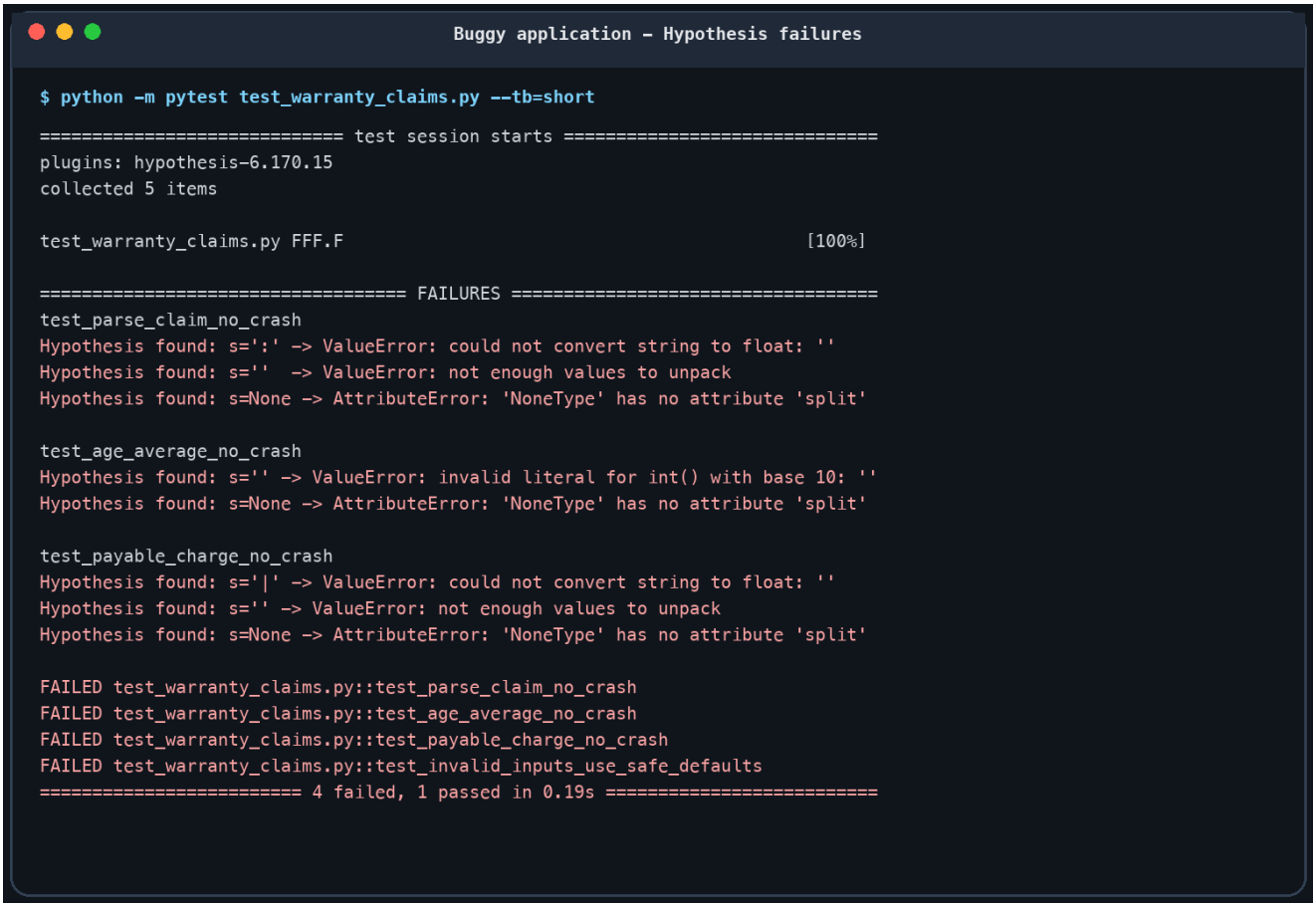
Step 2 - Original application output

Baseline run of the provided application with valid sample inputs.



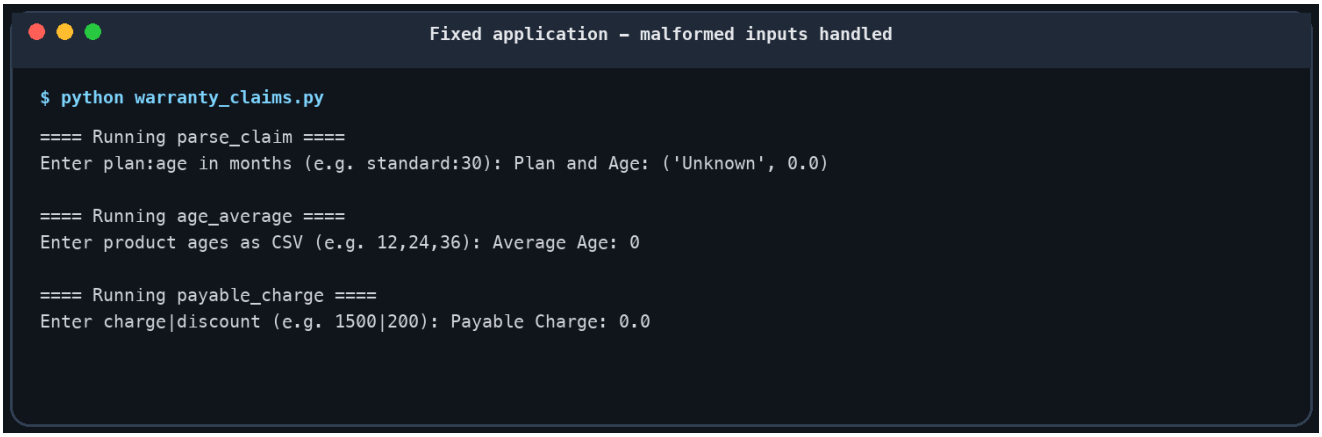
Step 4 - Failing fuzz tests

Hypothesis minimizes malformed inputs and reveals crashes in all three functions.



Step 5 - Fixed application output

The hardened application handles malformed values with safe defaults.



Step 6 - Passing fuzz tests

The final Hypothesis and regression test suite completes successfully.

